Finite Element Assembly

One-dimensional linear elements

What assembly must accomplish

For the axial bar,

$$\int_0^L AE\, u_h' v_h' \,\mathrm{d}x = \int_0^L f v_h \,\mathrm{d}x + \overline{P}\, v_h(L).$$

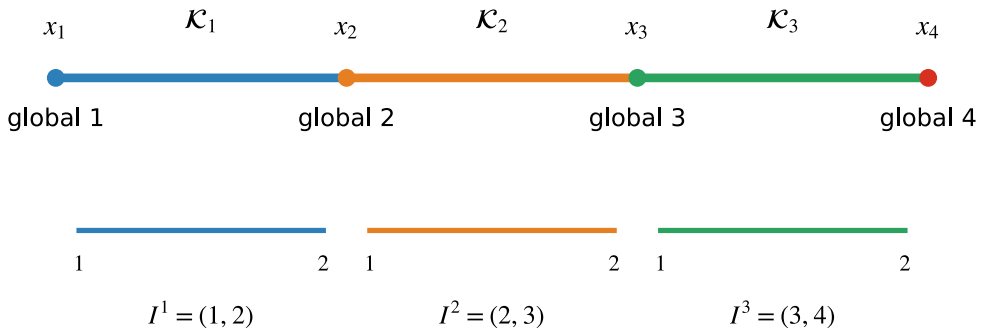
Each element supplies a small matrix and vector,

$$\boldsymbol{K}^e = \int_{\mathcal{K}_e} (\boldsymbol{B}^e)^T AE\, \boldsymbol{B}^e \,\mathrm{d}x, \qquad \boldsymbol{F}^e = \int_{\mathcal{K}_e} (\boldsymbol{N}^e)^T f \,\mathrm{d}x.$$

Assembly must place these local entries into the correct rows and columns of

$$\boldsymbol{K}\boldsymbol{U} = \boldsymbol{F}.$$

Local and global node numbers



Local node numbers restart on each element. Global node numbers identify shared nodes of the complete mesh.

Connectivity and incident elements

For a mesh with N_e two-node elements, $N_n = N_e + 1$ is the number of global nodes. The connectivity map is

$$I^e : \{1, 2\} \longrightarrow \{1, \ldots, N_n\}, \qquad I^e(1) = e, \quad I^e(2) = e + 1.$$

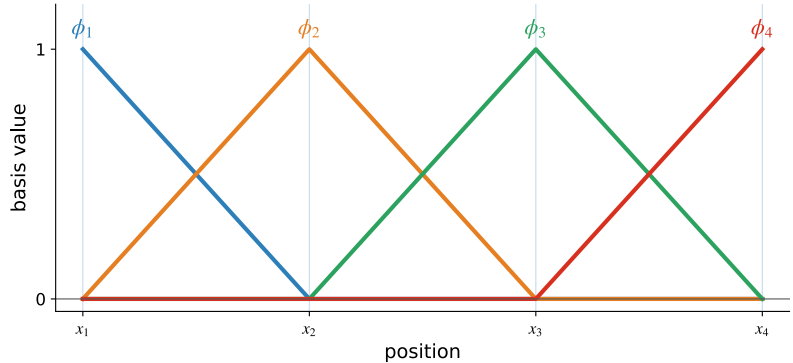
For a global node i that belongs to element e, $(I^e)^{-1}(i)$ gives its local number on that element.
The elements incident on global node i are

$$\mathcal{E}(i) = \{e : x_i \in \overline{\mathcal{K}}_e\}.$$

For the shared node 2,

$$J^1(2) = 2, \qquad J^2(2) = 1, \qquad \mathcal{E}(2) = \{1, 2\}.$$

Local support determines the interacting basis functions



$$\operatorname{supp} \phi_i = \bigcup_{e \in \mathcal{E}(i)} \overline{\mathcal{K}}_e.$$

On element e, only the basis functions associated with its two nodes are nonzero.

A global function restricted to one element

The global finite element function is

$$u_h(x) = \sum_{i=1}^{N_n} U_i \phi_i(x).$$

On $\mathcal{K}_e$, all terms except those associated with its two nodes vanish:

$$u_h|_{\mathcal{K}_e} = U_{I^e(1)} \phi_1^e + U_{I^e(2)} \phi_2^e = \boldsymbol{N}^e \boldsymbol{U}^e,$$

where

$$\boldsymbol{U}^e = \begin{bmatrix} U_{I^e(1)} \\ U_{I^e(2)} \end{bmatrix}.$$

Connectivity extracts the local coefficient vector from the global vector.

Load-vector assembly by global entry

The distributed part of global entry i is

$$F_i^{\mathrm{dist}} = \sum_{e=1}^{N_e} \int_{\mathcal{K}_e} f \phi_i \, \mathrm{d}x.$$

Only elements in $\mathcal{E}(i)$ contribute. Define

$$F_a^e = \int_{\mathcal{K}_e} f \phi_a^e \, \mathrm{d}x.$$

Since $\phi_i|_{\mathcal{K}_e} = \phi^e_{(I^e)^{-1}(i)}$,

$$\boxed{F_i^{\mathrm{dist}} = \sum_{e \in \mathcal{E}(i)} F^e_{(I^e)^{-1}(i)}.}$$

This formula fixes a global node i and gathers all incident-element contributions.

Three-element load vector

For

$$I^1 = (1,2), \qquad I^2 = (2,3), \qquad I^3 = (3,4),$$

the distributed-load vector is

$$\boldsymbol{F}^{\mathrm{dist}} = \begin{bmatrix} F_1^1 \\ F_2^1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ F_1^2 \\ F_2^2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ F_1^3 \\ F_2^3 \end{bmatrix} = \begin{bmatrix} F_1^1 \\ F_2^1 + F_1^2 \\ F_2^2 + F_1^3 \\ F_2^3 \end{bmatrix}.$$

At a shared node, the neighboring element contributions add.

Stiffness-matrix assembly by global entry

The global entry is

$$K_{ij} = \sum_{e=1}^{N_e} \int_{\mathcal{K}_e} AE\, \phi_j' \phi_i' \, \mathrm{d}x.$$

An element contributes only if it contains both nodes i and j. With

$$K_{ab}^e = \int_{\mathcal{K}_e} AE\, (\phi_b^e)'(\phi_a^e)' \, \mathrm{d}x,$$

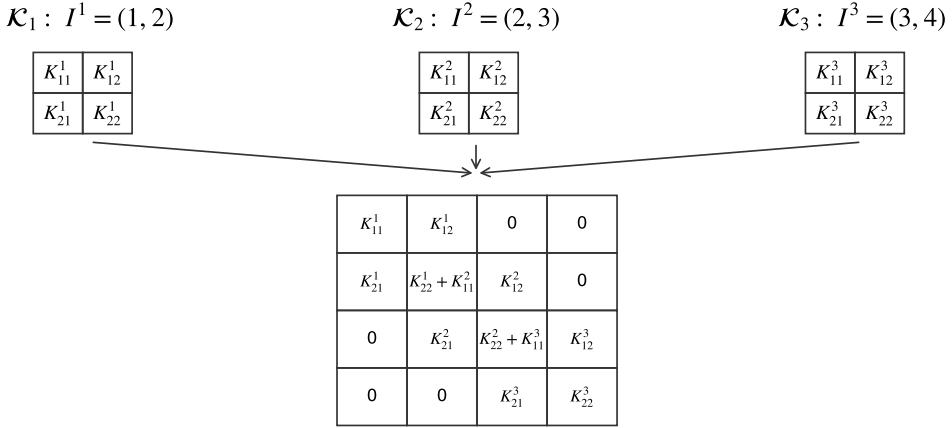
we obtain

$$\boxed{K_{ij} = \sum_{e \in \mathcal{E}(i) \cap \mathcal{E}(j)} K^e_{(I^e)^{-1}(i),(I^e)^{-1}(j)}.}$$

For example,

$$K_{22} = K_{22}^1 + K_{11}^2, \qquad K_{23} = K_{12}^2, \qquad K_{14} = 0.$$

Three-element stiffness matrix



shared entries accumulate contributions from adjacent elements

The assembly operation used in code

The entry formulas gather contributions for a fixed global index. A program usually loops over elements and scatters local entries:

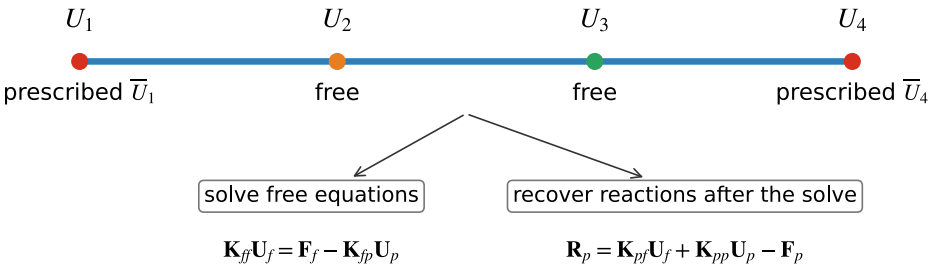
$$F_{I^e(a)} \mathrel{+}= F^e_a, \qquad K_{I^e(a),I^e(b)} \mathrel{+}= K^e_{ab}.$$

```
for e in range(num_elements):
    dofs = connectivity[e]
    K_e, F_e = element_arrays(e)

    for a in range(2):
        F[dofs[a]] += F_e[a]
        for b in range(2):
            K[dofs[a], dofs[b]] += K_e[a, b]
```

The arrays use zero-based Python indices. The mathematical map I^e uses one-based indices.

Essential boundary conditions



With free indices f and prescribed indices p,

$$\boldsymbol{K}_{ff}\boldsymbol{U}_f = \boldsymbol{F}_f - \boldsymbol{K}_{fp}\boldsymbol{U}_p.$$

Solve only for the free coefficients, then insert the prescribed values into the complete vector $\boldsymbol{U}$.

Reactions and final checks

After constructing the complete coefficient vector,

$$\boldsymbol{R} = \boldsymbol{K}\boldsymbol{U} - \boldsymbol{F}.$$

The entries associated with prescribed degrees of freedom are the reactions:

$$\boldsymbol{R}_p = \boldsymbol{K}_{pf}\boldsymbol{U}_f + \boldsymbol{K}_{pp}\boldsymbol{U}_p - \boldsymbol{F}_p.$$

Before accepting the solution, check:

- the connectivity and element lengths;
- the free and prescribed degree-of-freedom sets;
- global equilibrium using applied loads and reactions;
- displacement and axial-force trends against the loading.

Complete finite element procedure

1. Create node coordinates and element connectivity.
2. Initialize the global matrix $\boldsymbol{K}$ and vector $\boldsymbol{F}$.
3. For each element, compute $\boldsymbol{K}^e$ and $\boldsymbol{F}^e$.
4. Scatter-add the element entries using I^e.
5. Separate free and prescribed degrees of freedom.
6. Solve the reduced system for $\boldsymbol{U}_f$.
7. Insert prescribed values and compute reactions.
8. Evaluate displacement, strain, stress, and axial force elementwise.

Notebook demonstration problem

Set $L = AE = f_0 = 1$ and solve

$$-u'' = 1 \quad \text{in } (0,1), \qquad u(0) = 0, \qquad u'(1) = 0.$$

The exact solution is

$$u(x) = x - \frac{x^2}{2}.$$

During the demonstration:

- assemble and inspect $\boldsymbol{K}$ and $\boldsymbol{F}$ for one mesh;
- impose the essential boundary condition and solve;
- examine the displacement error; and
- then introduce derivative-based errors and mesh refinement.

Element loop in the demonstration

For uniform linear elements with $h = x_{e+1} - x_e$,

$$\boldsymbol{K}^e = \frac{1}{h}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad \boldsymbol{F}^e = \frac{h}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

```
for e in range(num_elements):
    dofs = np.array([e, e + 1])
    h_e = nodes[e + 1] - nodes[e]

    K_e = np.array([[1.0, -1.0],
                    [-1.0, 1.0]]) / h_e
    F_e = 0.5 * h_e * np.ones(2)

    for a in range(2):
        F[dofs[a]] += F_e[a]
        for b in range(2):
            K[dofs[a], dofs[b]] += K_e[a, b]
```

Solve, recover the reaction, and compare

```
num_nodes = len(F)
U = np.zeros(num_nodes)
K_ff = K[1:, 1:]
F_f = F[1:]
U[1:] = np.linalg.solve(K_ff, F_f)

reaction = (K @ U - F)[0]
```

For this problem, the computed nodal values agree with the exact quadratic solution at the nodes. Between nodes, u_h is piecewise linear. Mesh refinement reduces the interpolation error.

Begin with the displacement error

Let $e = u - u_h$. We first examine errors in the solution values:

$$\|e\|_{L^2} = \left(\int_0^1 e^2 \, \mathrm{d}x \right)^{1/2}, \qquad \|e\|_\infty = \max_{x \in [0,1]} |e(x)|.$$

For this problem, u_h agrees with u at the nodes. The exact solution is quadratic, while u_h is linear on each element. The error therefore vanishes at the nodes but is nonzero inside each element.

For linear elements in this particular problem,

$$\|e\|_{L^2} = O(h^2), \qquad \|e\|_\infty = O(h^2).$$

Energy and H^1 errors

The preceding norms do not measure the derivative error. For the bar problem,

$$\|e\|_E = \left(\int_0^1 AE(e')^2 \, \mathrm{d}x \right)^{1/2},$$

while

$$\|e\|_{H^1} = \left(\|e\|_{L^2}^2 + \|e'\|_{L^2}^2 \right)^{1/2}.$$

Here $AE = 1$, so the energy norm is the H^1 seminorm. Since $e(0) = 0$, the energy and H^1 norms are equivalent by the Poincaré inequality.
For linear elements and this smooth solution,

$$\|e\|_E = O(h), \qquad \|e\|_{H^1} = O(h).$$

Analysis in the notebook

For one mesh, the notebook displays:

- the finite element and exact solutions in the left panel;
- the pointwise absolute error in the right panel; and
- the L^2 and supremum errors in the error-plot title.

Only after interpreting these errors do we compute the energy and H^1 errors. The mesh-refinement loop then displays:

- all finite element solutions with the exact solution; and
- all four error norms against h on log–log axes.

For this problem, the fitted slopes should be approximately 2 for the L^2 and supremum errors and approximately 1 for the energy and H^1 errors.